

☒ Overview of Mobile Malware

☒ What is Mobile Malware?

Mobile malware is malicious software specifically designed to attack mobile devices like smartphones and tablets. These can steal data, track users, send unauthorized messages, or even control the device remotely.

☒ Common Types of Mobile Malware

1.

Trojans – Disguised as legitimate apps but perform malicious actions.

2.

Spyware – Monitors user activity and sends data to attackers.

3.

Adware – Floods device with ads; may collect personal info.

4.

Ransomware – Locks device or encrypts data demanding ransom.

5.

Worms – Spread via SMS, Bluetooth, or infected apps.

6.

Banking Trojans – Target financial apps and credentials.

7.

Rootkits – Gain root access and hide from security software.

☒ Infection Vectors

- Malicious apps (often sideloaded)
- SMS or email links
- Public Wi-Fi attacks
- Bluetooth or NFC exploits
- Fake app stores

☒ Common Malware Capabilities

- Keylogging
 - SMS interception
 - Data exfiltration
 - Screenshot capture
 - Command-and-control (C2) communication
 - Device location tracking
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☒ Android App Analysis

☒ What is Android App Analysis?

It's the process of reverse-engineering or examining Android applications (APK files) to understand their behavior, permissions, and potentially malicious activities.

☒ Key Tools

- APKTool – Decompile APK into readable resources (smali code).
- JADX – Convert APKs to Java source code.
- MobSF (Mobile Security Framework) – Static and dynamic analysis.
- Frida / Xposed – Dynamic instrumentation.
- Drozer – Attack framework for Android.

- **adb (Android Debug Bridge)** – Interact with Android device/emulator.

☒ Techniques Used

1.

Static Analysis – Analyzing code without execution.

- Disassembling with JADX or APKTool.
- Reviewing AndroidManifest.xml for permissions and components.

2.

Dynamic Analysis – Running the app in a controlled environment.

- Monitoring file access, API calls, network traffic.

3.

Behavioral Analysis – Observing how the app behaves on a real/emulated device.

☒ **Key Focus Areas**

- Requested permissions vs actual behavior.
- Network connections (to C2 servers).
- Embedded payloads or obfuscation.
- Sensitive data handling and leakage.
- Runtime behavior (e.g., screenshotting, recording).